

porting. The idea struck Dr Arjun Kalyanpur while working at the Emergency Room at the Yale School of Medicine, US. "During night shifts, we accessed scans from a hospital across town which were transmitted electronically to us," he says. "I began to think that the same thing could be done from India, where it would be daytime." But the idea of starting a company came later when he met a radiologist friend in the US by chance. When he told his wife, Sunita Maheshwari, also a Yale medic, her response was instant: "That is a space-age concept." But the couple anyway put down their savings and started working from home through the Internet for US clients. Now with the Indian health imaging market expected to double from the existing Rs 1,575 crore in the next five years, competition is brewing up. Wipro, Reliance and Apollo have recently joined the fray. But Telerad holds 90 per cent of the market share in the country, growing at 50 per cent year on year.

Driven by the rising healthcare demands and spending power of India's affluent generation, medical technology looks set to enter a golden age. A new FICCI-Ernst & Young study predicts 15-20 per cent growth for the Indian medical equipment market, slated to grow from Rs 9,000 crore now to Rs 22,500 crore by 2012. Not surprisingly, private hospitals are taking the lead in introducing the latest technological wonders and creating milestones in treatment. In 2006, Asia's first high field strength in-

traoperative MRI (iMRI), the most advanced technology to treat brain tumours with utmost precision, was launched at the Institute of Neuroscience of the Max chain in Delhi. In the past, the difficulty in distinguishing between diseased and healthy brain tissue was addressed with a follow-up MRI scheduled a day or so

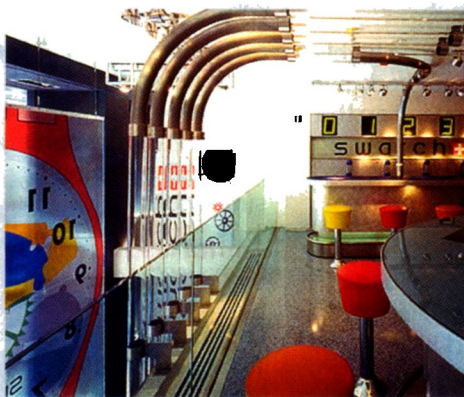
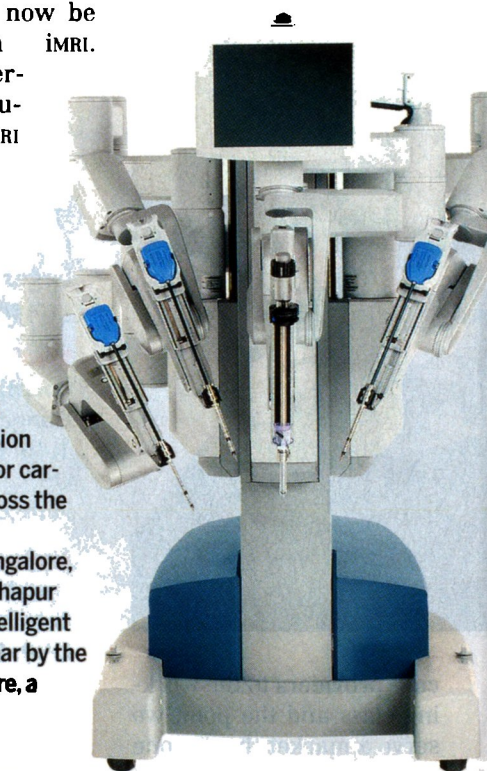
after the surgery. In cases where residual tumour remained, a decision had to be made about whether another operation was a viable option. That delay between surgery and follow-up MRI could now be eliminated with iMRI. Surgeons could operate on a brain tumour, slide the MRI

into the surgical suite for a scan and immediately assess whether more surgery is needed. By 2010, a host of private hospitals—from Medanta to Kokilaben Dhirubhai Ambani Hospital

New Technologies

Robotic Surgery

A mechanical unit with arms and tiny hand-like instruments that the surgeon controls with joystick-like devices from a console. The small incisions and extreme precision make it a patient-friendly procedure. Was introduced for cardiac surgery by Escorts in 2001. Today used widely across the country: Jaslok and Hiranandani hospitals in Mumbai, Narayana Hrudayalaya and Wockhardt Hospitals in Bangalore, Care hospital in Hyderabad etc. Swarup Hospital in Kolhapur has an indigenous method of robotic laparoscopy; "intelligent robotics" for cancer—CyberKnife—was brought last year by the Apollo, Chennai. A Da Vinci robot costs about Rs 10 crore, a cyberknife system 65-70 crore.



Pneumatic Chutes

They connect and serve hospital departments, transporting pharmaceuticals, lab samples and sensitive medical items at high speed. Done manually once, it ensures integrity of diagnostic processes, patient safety and service efficiency. Almost every specialty hospital in India has introduced the chute.

Paperless, Filmless

Scribbled notes, bills, prescriptions, large diagnostic charts, reports and even patient queues will soon be a thing of the past. Hospital workflows—from patient consultation, prescription, investigations, registration, doctor and nursing notes, billing, inventory management to discharge—are increasingly just a click away. Modern hospitals are now starting out with hospital information systems in place. Artemis Health Institute of Gurgaon, for instance, had an IT budget of Rs 6.5 crore when it started in 2007.

